

and the other at the front for video conferencing. This adds to the price of the phone.

The applications based on 3G are either provided by the service provider or bundled with the handset by the manufacturer. Most 3G handsets come with inherent software tools required for 3G applications. If you intend to use a particular application and you do not have it on your 3G phone, you may download it from the manufacturer or service provider's site. For example, if you do not have the video conferencing software, you may need to download it. But if your media player is capable of streaming a video provided by the telecom operator, you may view live TV directly without downloading any other software. For watching a video online, you will require Flash memory and a mobile Internet browser.

Interoperability and backward compatibility

There is a lot of confusion about interoperability and backward compatibility when we talk about moving from 2G world to 3G world. If you buy a 3G mobile phone that fascinates you, but you do not have a 3G network in your area, it will not be of any use. You will be able to do only what you could do with your 2G phone. You need a 3G network to be able to receive 3G content and services. Keep in mind that not all networks are created equally. So if you are planning to buy a 3G cellphone, make sure that the operator offers 3G network and service in your area.

Service operators provide you with a separate 3G SIM card to use with the 3G phone. It's only with a 3G-enabled SIM that you can enjoy 3G services. "You can use a 2G SIM card in 3G phone without any problem. You will be able to use all the features of 2G and 2.5G but not the 3G features," explains Anil Kaushik, vice president-executive, Fly Mobile. You can also use a 3G SIM in your 2G handset, but it will allow you to use only 2G features. To avail

Tasks at hand

"We are still in the world of DOS with mobile world," rightly said Sangeet Chowla, chief strategy officer, Comviva, at IMAI's Mobile Content and Services Conference. There are several challenges in terms of use of technology.

Most low-cost handsets cannot run high-end applications. The solution suggested by Chowla includes remote applications by means of mobile cloud technology. In this technology, the complexity of the application is pushed to the network rather than the handset.

Another big challenge is too many mobile platforms. Mobile phones are available on a variety of operating systems such as Windows Mobile, Android, Symbian, RIM OS and Apple OS, and they use different mobile Web browsers. Embedded application developers need to make use of cross-platform or multi-platform tools so that their application is independent of the mobile handset.

Mobile Web faces the limitation of screen size and keypad—the two handset features that play an important role for many applications. Content on the Web needs to be formatted to fit the screen size of the handset and make scrolling easy. The application too needs to be user-friendly.

Chowla shares, "The next-generation applications are driven by data. Data plans are enemy of data applications as users do not understand whether 1 kbps of data permits receiving e-mail or watching a video." It's easier for users to understand one rupee per SMS. Conveying technical information simply so that a common man can understand it poses another big challenge.

A research shows that there are 37 per cent English literates in urban India. Going by this number, there is a need to develop regional language content and related audio-visual applications.

3G features, you need a 3G mobile handset with a 3G SIM card.

Varied base technologies

Although it does not make a recognisable difference to people who use 3G for fun, various 3G technologies play an important role in providing high-speed data. "In CDMA, 3G is provided by evolution-data-optimised (EVDO) technology. EVDO provides improvised data rates from revision 0 (Rev 0) to revision A (Rev A) and then revision B (Rev B). As of now, all the operators in CDMA space have EVDO Rev A technology," informs Kaushik. EVDO Rev A provides maximum download speed of 3.1 Mbps and uplink speed of 1.8 Mbps.

About GSM standards (for 3G it is 3GSM or UMTS), he says, "GSM has adopted WCDMA technology for 3G. HSDPA and HSUPA are the other high-speed transmission technologies accepted by GSM standards." A speed of 384 kbps can be achieved with WCDMA technology, but with high-speed downlink packet access (HSDPA) the downlink speed can be as high as 14.0 Mbps. Currently, 3G handsets are either based on WCDMA

or HSDPA technology.

What's on offer?

We have been talking a lot about 3G offering high-speed Internet and more and more data-based applications. But with 3G you can have better-quality service for voice calls too. 3G technologies will offer efficient spectrum usage to accommodate more and more users.

"As more and more urban people move from 2G to 3G, the vacated space in 2G will give a way to increase rural telephony," explains Kaushik. The operators would like to add more customers to the freed-up 2G spectrum. "As 2G is an already established network, the cost of 2G services is expected to come down. This will also increase the Internet penetration via 2G to the rural India," foresees Kaushik.

Talking about availability of the 3G network, Soni adds, "As of now, you only have BSNL and MTNL providing 3G services. When the auction happens, and when bigger operators start operating 3G services, the real story of 3G will start for majority." ●

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